Vinyl Crate Dashboard: Design Document

Overview

A self-hosted personal dashboard styled as a vintage record store interface, where server services, system statistics, and homelab information are presented as vinyl records in a browseable crate. The aesthetic draws from warm analog audio equipment and record store culture, creating an engaging and unique way to monitor infrastructure while celebrating the user's audio equipment interests.

Core Concept

The dashboard metaphor centers around a vinyl record crate or shelf. Each "record" represents a different service, system metric, or data source from the homelab environment. Users can "flip through" records to view different information, with each record showing relevant stats, status indicators, and quick actions on its album artwork and liner notes.

Visual Design Philosophy

Color Palette

The design should evoke warm, analog aesthetics associated with vintage audio equipment and record stores:

- Warm cream and off-white backgrounds (aged paper quality)
- Rich walnut and dark wood tones for structural elements
- Muted orange, amber, and copper accents (VU meter glow, warm lighting)
- Deep blacks for text and high-contrast elements
- Occasional pops of faded album art colors (blues, reds, yellows from classic album covers)

Typography

- Display fonts: Retro sans-serifs or geometric fonts reminiscent of 1960s-70s album typography
- Body text: Clean, readable sans-serif for data and metrics
- Monospace font for technical stats (IP addresses, uptime counters)
- Hand-drawn or script fonts sparingly for "handwritten" labels or notes

Textures and Materials

- Subtle paper grain or canvas texture on backgrounds
- Wood grain on structural UI elements (crate dividers, shelves)
- Vinyl record texture with realistic grooves and label circles
- Cardboard texture for record sleeves
- Slight wear and aging effects (scuffs, dust, ring wear on records)

User Interface Components

The Record Crate

The primary navigation interface. Records are displayed edge-on in a horizontal scrollable crate, showing spine labels with service names. Users can click or scroll through records to browse their infrastructure.

Features:

- Horizontal scrolling with smooth momentum physics
- Divider cards to organize records by category (Services, VMs, Network, Media, etc.)
- Visual depth with subtle 3D perspective on the crate
- Mouse wheel or trackpad scroll to flip through records
- Keyboard arrow keys for navigation

Individual Record Cards

When a record is selected, it expands to show the full album artwork view with detailed information overlaid or on a flip side.

Front (Album Art):

- Service icon or custom artwork styled as album cover
- Service name as "artist" and "album title"
- Color-coded corner or edge based on service status (green for healthy, red for down, amber for warnings)
- Small indicators for quick stats (CPU usage as a small gauge, memory as a progress ring)

Back (Liner Notes):

- Detailed metrics presented as track listing
- Uptime shown as "runtime" or "total play time"
- Resource usage broken down as individual "tracks"
- Quick action buttons styled as record label stamps or stickers
- Links to service UIs presented as "credits" or "produced by" information

The Turntable (Now Playing Section)

A dedicated area showing currently active or most important service, styled as a record player with spinning vinyl.

Features:

- Animated spinning record when service is active
- Tonearm that moves based on activity levels or throughput
- VU meter style gauges for CPU and RAM usage
- Soft glow effects around the platter
- Could integrate with Home Assistant to show actual audio playback if music is playing

Shelf or Wall Display

Background or header area showcasing overall system health and ambient information:

- "Shelf" of record spines showing all services at a glance
- Vintage analog clock for current time
- Weather widget styled as album art or poster
- Network activity shown as LED indicators or small VU meters
- System temperature gauges styled as analog thermometers

Data Sources and Integration

Homelab Services

The dashboard should pull data from Mason's existing infrastructure:

- **Proxmox**: VM status, resource allocation, node health
- **Home Assistant**: Device states, automation status, sensor data
- **UniFi Network**: Client count, bandwidth usage, AP status
- **Game Servers**: Minecraft server status, player counts, uptime
- **Docker/Container Stats**: Running containers, resource usage per service
- **System Metrics**: CPU, RAM, disk usage, temperatures
- **Network Services**: Traefik reverse proxy status, SSL certificate expiry

API Integration Requirements

- RESTful endpoints for real-time data fetching
- WebSocket or Server-Sent Events for live updates without polling
- Authentication layer for secure access (OIDC integration via existing setup)
- Caching layer to prevent overwhelming source APIs
- Graceful degradation when services are unavailable

Interaction Design

Navigation Patterns

1. **Browse Mode**: Scroll horizontally through the record crate, clicking records to expand
2. **Detail View**: Click a record to flip it over and see detailed stats and actions
3. **Quick Actions**: Hover over record spines to see quick status tooltips
4. **Search/Filter**: Type-to-search functionality to quickly find specific services
5. **Categorization**: Organize records into crates or sections by type

Animations and Transitions

- Smooth record pull-out animation when selecting
- Flip animation to transition between album art and liner notes
- Vinyl spin animation with realistic physics (inertia, slowdown)
- VU meter needle bounce responding to activity
- Subtle parallax effects on background elements
- Dust particles or ambient particles for atmosphere

Responsive Behavior

- Desktop: Full crate view with turntable section
- Tablet: Simplified crate with vertical scrolling option
- Mobile: Stack view showing records as cards in a list
- Adaptive layouts that maintain the aesthetic across screen sizes

Technical Considerations

Performance

- Canvas or WebGL for record animations if needed
- CSS transforms for smooth 60fps animations
- Lazy loading for record artwork and data
- Efficient DOM updates (virtual scrolling for large collections)
- Image optimization for album art assets

Accessibility

- Keyboard navigation support (tab through records, arrow keys to scroll)
- Screen reader labels for all interactive elements
- High contrast mode option that maintains aesthetic
- Focus indicators styled to match vintage theme
- Alternative text descriptions for visual information

Customization

- User-configurable record order and categorization
- Custom album artwork upload for services
- Adjustable color schemes (warm/cool variants of the palette)
- Toggle for animation intensity (performance vs. aesthetics)
- Widget visibility controls (show/hide turntable, shelf, etc.)

Content Strategy

Record Artwork Generation

Options for creating album art for each service:

1. Use service logos directly, styled with vintage filters
2. Generate procedural artwork based on service type and colors
3. Allow custom image uploads
4. Create themed artwork sets (all services as fictional band albums)
5. Pull from existing icon libraries with vintage post-processing

Service Categorization

Suggested "crate dividers" or sections:

- **Infrastructure**: Proxmox nodes, network equipment, core services
- **Media**: Plex, music servers, streaming services
- **Gaming**: Minecraft servers, game infrastructure
- **Home**: Home Assistant, smart home devices, automation
- **Development**: Git servers, CI/CD pipelines, dev environments
- **Monitoring**: Grafana, uptime monitors, log aggregation

Data Visualization on Records

Creative ways to show metrics within the vinyl/album metaphor:

- CPU usage as concentric rings on the vinyl label
- Memory usage as a progress ring around the record edge
- Network throughput as VU meter bounces
- Uptime as "plays since release date"
- Disk space as "remaining tracks" or "B-side availability"
- Service version as "pressing edition" or "catalog number"

Mood and Atmosphere

Audio Design (Optional Enhancement)

While not required, subtle audio cues could enhance the experience:

- Soft vinyl crackle ambient sound
- Gentle click when flipping through records
- Notification sounds styled as tape rewind or needle drop
- Background music option (user's own library via HA integration)

Lighting and Ambiance

- Warm ambient lighting simulation (as if lit by tungsten bulbs)
- Subtle shadows and depth to create dimensional space
- Glow effects on active elements (turntable light, VU meters)
- Time-of-day theming (warmer at night, brighter during day)

Use Cases

Primary Use Case: Infrastructure Monitoring

Mason checks the dashboard multiple times daily to:

- Verify all critical services are running
- Monitor resource usage across Proxmox VMs
- Check network client counts and bandwidth
- Ensure game servers are accessible
- Spot any alerts or anomalies requiring attention

Secondary Use Case: Visual Display

The dashboard serves as an ambient display on a monitor or wall screen:

- Aesthetic enough to leave visible in living spaces
- Informative at a glance without demanding attention
- Conversation piece that showcases homelab in stylish way
- Reflects personal interests (audio equipment, DIY culture)

Tertiary Use Case: Quick Actions

From the dashboard, Mason can:

- Launch service web interfaces
- Trigger Home Assistant automations
- Restart services or containers
- Acknowledge alerts or notifications
- Access service logs or documentation

Future Expansion Ideas

Phase 2 Enhancements

- RSS feed integration shown as "new releases" section
- Weather and calendar widgets styled as album inserts or posters
- Notification system using "listening booth" or "notice board" metaphor
- Integration with actual audio playback (vinyl icon spins when music plays)
- Historical data visualization as "discography" or "collection timeline"

Advanced Features

- Collaborative features (other users can "browse the collection")
- Mobile companion app with notification support
- Voice control integration ("play the Proxmox record")
- AR mode to project dashboard into physical space
- Export dashboard state as shareable "mixtape" snapshot

Success Metrics

The dashboard will be considered successful if it:

1. Provides at-a-glance infrastructure health visibility
2. Remains performant across Mason's devices and browsers
3. Feels engaging and pleasant to interact with regularly
4. Successfully integrates with existing homelab APIs
5. Maintains aesthetic coherence while being functional
6. Becomes the preferred way to check homelab status over individual service UIs

Conclusion

The Vinyl Crate Dashboard transforms infrastructure monitoring from a utilitarian task into an aesthetically rich experience that aligns with Mason's interests in audio equipment and DIY culture. By leveraging the familiar and beloved metaphor of vinyl records and record stores, the dashboard creates an emotional connection to the homelab while providing all necessary technical functionality. The warm, analog aesthetic offers a refreshing counterpoint to typical modern dashboard designs, making it both a practical tool and an expression of personal style.